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National Chemical Laboratory Campus, Pune 4 DAY-WORKSHOP ON MULTI-CORE COMPUTING (Introduction to OpenCL, MPI, MMX & OpenCL)

FACULTY:

S. Basu, MTech (IIT-KGP),
ex-Intel, exp in VLSI design

For: Signal & Image processing researchers;
Entrepreneurs; PG/final yr UG students with
Computer Sci. background; all IT Professionals

Dates: 4/11/18/25 August 2012 | 10am – 5pm

Venue: E-Classroom, NCL Innovation Park,
Pashan Road, Pune
www.venturecenter.co.in/workshops
20250934/64011026,
Email: eventsdesk@venturecenter.co.in

Upcoming events at Venture Center

September '12: Embedded Systems for Robotics Part-I
October '12: Embedded Systems for Robotics Part-II
November '12: Digital Design using Verilog



From Facebook: On PCBs

I am facing problems in designing a double-sided PCB for the microprocessor training kit. It is a very complex circuit. Any solution for that? I am working on ExpressPCB software.

Krishan M. Kaushik

☐ ExpressPCB is easy to use but it does not give you the Gerber file. It will be a pain if your PCB maker accepts only Gerber format (the same happened with me a few months ago). So better try Eagle or Protel software. These are industry-standard tools and very, very powerful.

Saikat Das Adhikari

☐ In construction projects of EFY magazine, the board layout is mentioned as 'actual size' but how can we fabricate exact-size PCBs from it? We have to move it to some kind of glossy paper. For that, we have to photocopy it. Will the size not vary then? I don't think we have to make a mirror image.

Siddharth Chandrasekaran

☐ Yes, sometimes the print size of the photo-copy differs slightly but for most of the through-hole PCB layout it is not much of a problem. Also, the track layout given in the magazine is sufficiently thick and minor change in size can be adjusted during drilling. If the layout is scanned, you may get a slight change in the size of the printed layout. An interesting observation: Download a layout from the Internet thrice and print the downloads using a laser printer. Cut the three layout prints and compare their size against a light source. You are likely to find a slight difference in the size.

If you print the layout from the PDF by unchecking the 'Fit to page' option and compare the printout with the original (published), then also you may observe a slight difference. I have experienced this problem.

Vivek Ghosh

☐ Vivek, I didn't cross-check but all the components can be placed on the PCB without any problem. Thanks for your information.

Arun Sarma

☐ You simply xerox the layout from the magazine on a glossy paper by selecting the size to 100 per cent. The size will not vary at all. Next, transfer the toner to copper cladding using an electric iron. Your PCB is now ready for etching.

Vaibhav Srivastava

The author Petre Tsv Petrov replies:

You cannot replace TDA2005M with LM1875 in this circuit. LM1875 is a different chip and you need a different solution. Also, the current and power comparison you have made is not correct. With LM1875, you have the following options:

1. Guitar and microphone amplifier with single LM1875
2. Guitar and microphone amplifier with dual LM1875 in parallel
3. Guitar and microphone amplifier with dual LM1875 in bridge

Of these options, the first is the simplest and possibly the best if you do not have experience in the field.

If you are interested in any of these solutions, I can design the LM1875-

based circuit and send it to EFY for publication. But, you should specify either single or bipolar power supply for the LM1875.

Fastest Finger First Indicator

I am interested in assembling an 89C51-based fastest-finger first indicator with seven-segment display. Where can I get the circuit diagram and source code for it?

Mohammed Rafiq Munshi
Belgaum

EFY: A circuit for the same was published in June 2006 issue of EFY. You may purchase the circuit from www.efymagonline.com. The source code can be downloaded from www.efymag.com free of cost.